

Abhishek Rajesh Kharat

abhikharat0424@gmail.com • +91 9881164074 • Pune, Maharashtra
[GitHub](#) • [LinkedIn](#) • [Medium Blog](#) • [Hugging Face](#)

SUMMARY

Pre-final year engineering student pivoting into AI/ML and cloud. Currently interning as an Agentic AI Intern at Innomatics Research Labs, working with LangChain, LangGraph, FastAPI, and AWS. Independently built and deployed multiple AI-driven projects — from a Reinforcement Learning disaster-response environment to a CAD validation tool using ML — and published technical writing on NLP pipelines. Actively seeking paid AI/ML or data engineering internships.

EXPERIENCE

Agentic AI Intern • [Innomatics Research Labs](#) • *Remote* Feb 2026 – Present

- ▶ Working on **LangChain** and **LangGraph** agent pipelines for multi-step AI task automation.
- ▶ Building **FastAPI** backend services and integrating them with **AWS** cloud infrastructure.
- ▶ Published a technical blog on building an NLP preprocessing pipeline (narrative-driven, with custom SVG visuals) on Medium as part of internship assignments.
- ▶ Completed structured training in Python programming, Generative AI, and agentic system design.

PROJECTS

Disaster Response Coordination RL Environment — *Python · Gymnasium · PyTorch · Hugging Face Spaces*
[\[GitHub\]](#)

- ▶ Built a custom **Reinforcement Learning environment** from scratch for the **Meta/PyTorch OpenEnv Hackathon** — first-ever RL project, completed independently.
- ▶ Modelled multi-agent disaster coordination (resource allocation, routing) as a Gym-compatible environment with structured observation and action spaces.
- ▶ Deployed live on **Hugging Face Spaces** (AbhishekKharat11) with full documentation.

DesignGuard AI – Automated CAD Validation Tool — *Python · ML · Flask · PDF Reporting* [\[GitHub\]](#)

- ▶ Designed an **open-source CAD validation tool** for the Varroc Eureka Challenge 3.0 (solo, team NeuralBlue).
- ▶ Implemented CAD file parsing, ML-based defect detection, a Flask dashboard, and automated PDF report generation.
- ▶ Scope covers tolerances, material checks, and structural flag detection — aimed at reducing manual design review time.

Hood Reviews – Hyperlocal Neighbourhood Review App — *React · Google Maps API · Firebase · Vercel*
[\[GitHub\]](#)

- ▶ Built and deployed a full-stack **React web app** for neighbourhood reviews with global postal code support and Google Maps integration.
- ▶ Supports image/video upload, 35+ review categories, and live Vercel deployment.

FastAPI Hotel Booking API — *Python · FastAPI · REST* [\[GitHub\]](#)

- ▶ Developed a **production-style REST API** for hotel booking with CRUD operations, data validation using Pydantic, and modular route structure.

NLP Preprocessing Pipeline — *Python · NLTK · spaCy · Medium Blog*

- ▶ Implemented a full NLP pipeline (tokenisation → stopword removal → stemming/lemmatisation → vectorisation) and explained it through a narrative blog post with custom SVG visuals published on Medium.

SKILLS

Languages: Python, C++ (basic), Bash (basic)

AI / ML: LangChain, LangGraph, Gymnasium (RL), NLTK, spaCy, Prompt Engineering

Cloud & DevOps: AWS (Bedrock, QuickSight), Hugging Face Spaces, Vercel, Git, GitHub

Web & APIs: FastAPI, Flask, React (basic), REST APIs

Tools: Linux/Ubuntu, Jupyter, VS Code, Arduino IDE, MATLAB, Fusion 360

Concepts: Reinforcement Learning, NLP Pipelines, Agentic AI, REST API Design

EDUCATION

B.E. Automation & Robotics

Expected May 2027

Modern Education Society's Wadia College of Engineering, Pune

CERTIFICATIONS & ACHIEVEMENTS

- ▶ ISRO – Artificial Intelligence & Machine Learning
- ▶ Python Programming Certificate
- ▶ Technical Author on Medium — published NLP pipeline article
- ▶ German Language – A2 Beginner